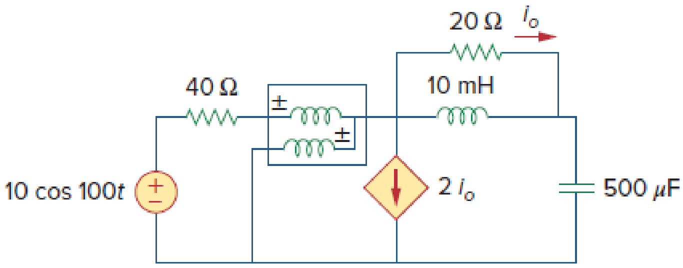
Chapter 11, Problem 79P

Problem

Determine the wattmeter reading of the circuit in Fig. 11.94.

Figure 11.94:



Step-by-step solution

Step 1 of 6

The wattmeter reads the power supplied by the source & partly by the 40 Ω resistor.

$\omega = 100$ red/sec

$100\,\mathrm{mH} \Rightarrow j100\times 10\times 10^{-3}$

$100\,\mathrm{mH} = j\ \Omega$

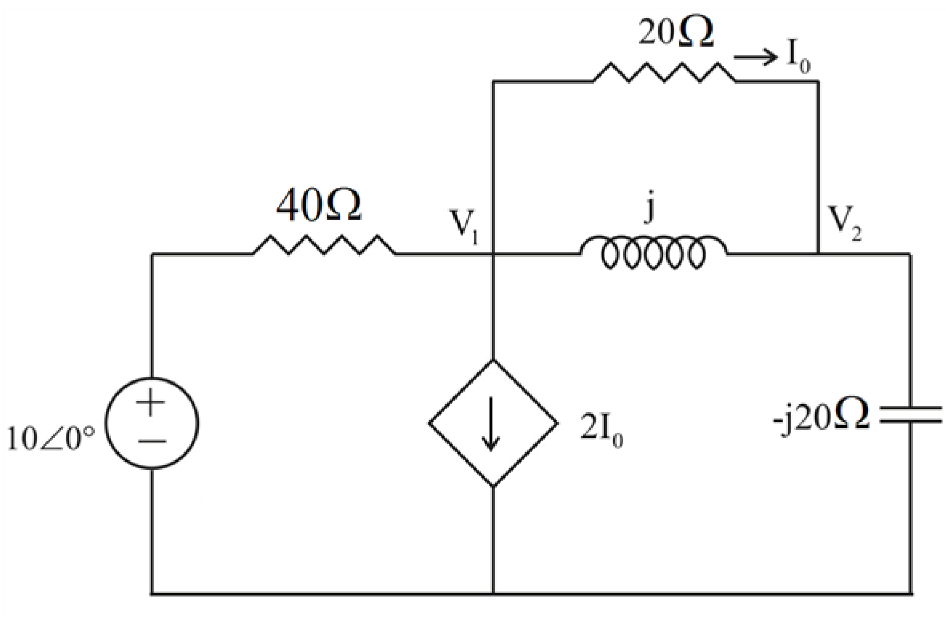
$$500\,\mu\mathrm{F} = \frac{1}{j\omega C}$$

$$500\,\mu\mathrm{F} = \frac{1}{j100\times 500\times 10^{-6}}$$

$500\,\mu\mathrm{F} = -j20\ \Omega$

Step 2 of 6

The frequency domain circuit is,



Step 3 of 6

In the above circuit,

$$I_0 = \frac{V_1 - V_2}{20} \text{ .... } (1)$$

Applying KCL to the above circuit we get,

At node 1,

$$\frac{10 - V_1}{40} = 2I_0 + \frac{V_1 - V_2}{j} + \frac{V_1 - V_2}{20}$$

$$\frac{10 - V_1}{40} = \frac{3(V_1 - V_2)}{20} + \frac{V_1 - V_2}{j} \quad (\text{from} - (1))$$

$$\frac{10 - V_1}{40} = \frac{3(V_1 - V_2)}{20} - j(V_1 - V_2)$$

$$10 - V_1 = 6(V_1 - V_2) - j40(V_1 - V_2)$$

$$10 = (7 - j40)V_1 + (-6 + j40)V_2 \text{ .... } (2)$$

Step 4 of 6

At node 2,

$$\frac{V_1 - V_2}{j} + \frac{V_1 - V_2}{20} = \frac{V_2}{-j20}$$

$$(V_1 - V_2) + \frac{j(V_1 - V_2)}{20} = -\frac{V_2}{20}$$

$$20(V_1 - V_2) + j(V_1 - V_2) = -V_2$$

$$0 = (20 + j)V_1 - (19 + j)V_2 \text{ .... } (3)$$

Step 5 of 6

Solving (2) & (3) yields,

$$V_1 = 1.5568 - j4.1405$$

$$V_1 = 4.424\angle -69.39^\circ$$

$$I = \frac{10 - V_1}{40}$$

$$I = \frac{10 - (1.5568 - j4.1405)}{40}$$

$$I = \frac{8.4432 + j4.1405}{40}$$

$$I = 0.21108 + j0.1035125$$

Step 6 of 6

$$S = \frac{1}{2}V_1 I^*$$

$$S = \frac{1}{2} \times (1.5568 - j4.1405) \times (0.21108 - j0.1035125)$$

$$S = \frac{-0.0999 + j0.7572}{2}$$

$$P = \text{Re}(S) = \boxed{0.04995 \text{ W}}$$